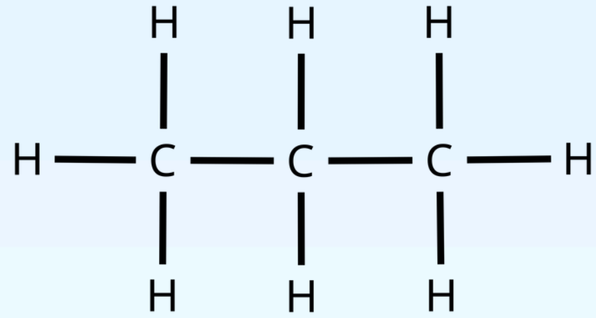
1-PROPANE (C_3H_8)

1.Molecular Formula

C_3H_8

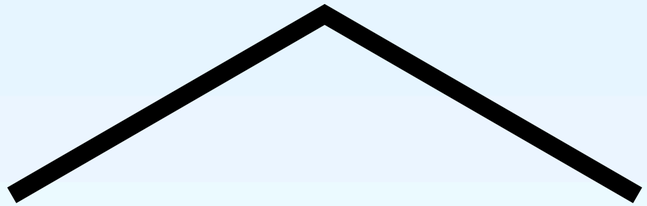
2.Lewis Structure



3.Condensed Structural Formula

$CH_3CH_2CH_3$

4.Bond-Line Formula



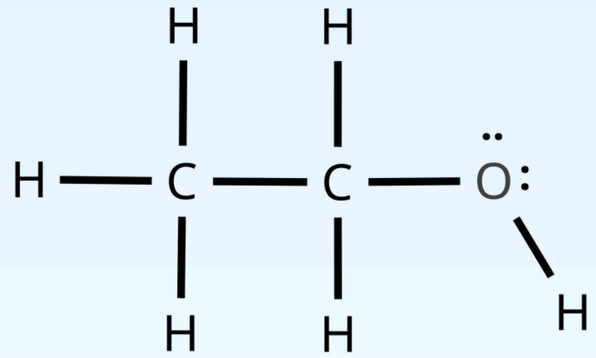
Atom count verified: 3 C, 8 H
Valence checked: each C has 4 bonds, each H has 1
Same connectivity confirmed: $CH_3CH_2CH_3$

2-ETHANOL (C_2H_6O)

1.Molecular Formula

C_2H_6O

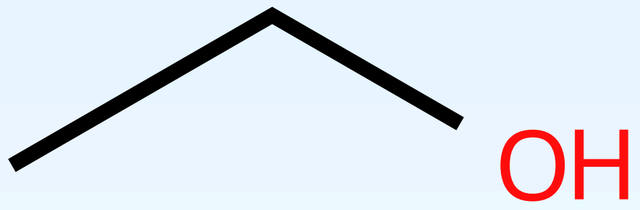
2.Lewis Structure



3.Condensed Structural Formula

CH_3CH_2OH

4.Bond-Line Formula



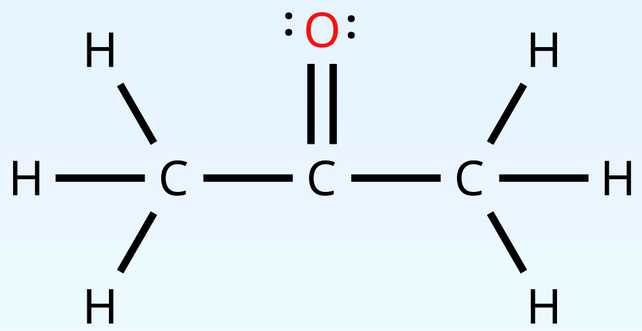
Atom count verified: 2 C, 6 H, 1 O
Valence checked: C 4 bonds, O 2 bonds, + 2 lone pairs
Same connectivity confirmed: CH_3CH_2OH

3-ACETONE (PROPAN-2-ONE)

1.Molecular Formula

C_3H_6O

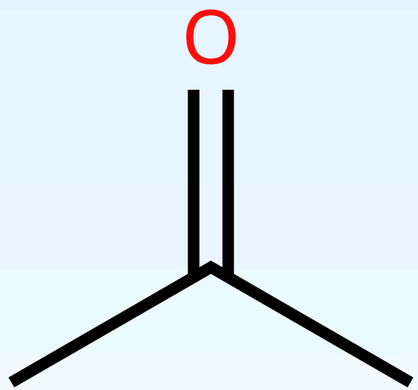
2.Lewis Structure



3.Condensed Structural Formula

$CH_3C(=O)CH_3$

4.Bond-Line Formula



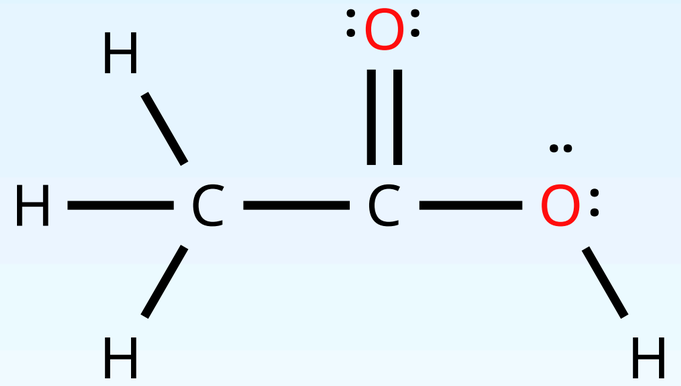
Atom count verified: 3 C, 6 H, 1 O
Valence checked: C=o counts as 2 bonds, O has 2 lone pairs
Same connectivity confirmed: $CH_3C(=O)CH_3$

4-ACETIC ACID (ETHANOIC ACID)

1.Molecular Formula

$C_2H_4O_2$

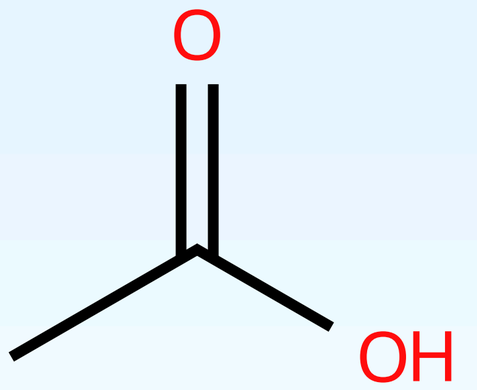
2.Lewis Structure



3.Condensed Structural Formula

$CH_3C(=O)OH$

4.Bond-Line Formula



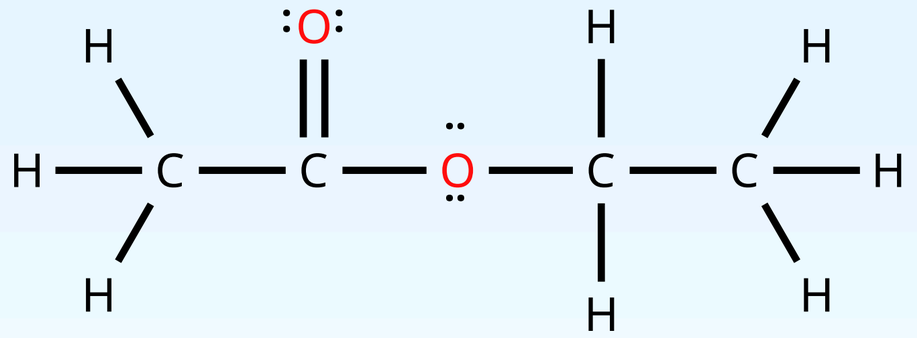
Atom count verified: 2 C, 4 H, 2 O
Valence checked: 1 C=O, 1 O-H, each O has 2 lone pairs
Same connectivity confirmed: $CH_3C(=O)OH$

5-ETHYL ACETATE (ETHYL ETHANOATE)

1.Molecular Formula

$$C_4H_8O_2$$

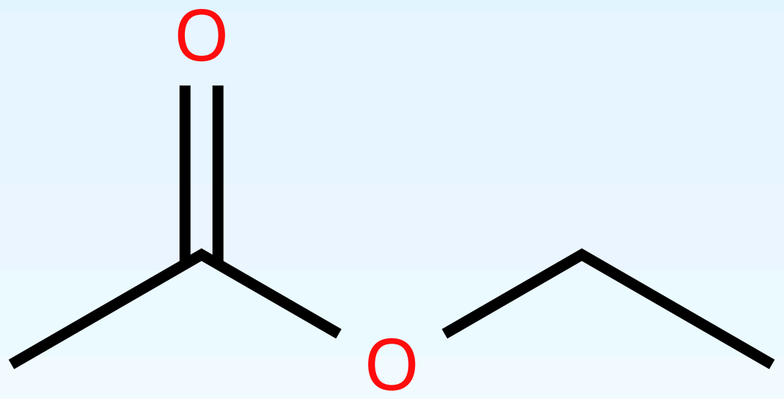
2.Lewis Structure



3.Condensed Structural Formula

$$CH_3C(=O)OCH_2CH_3$$

4.Bond-Line Formula



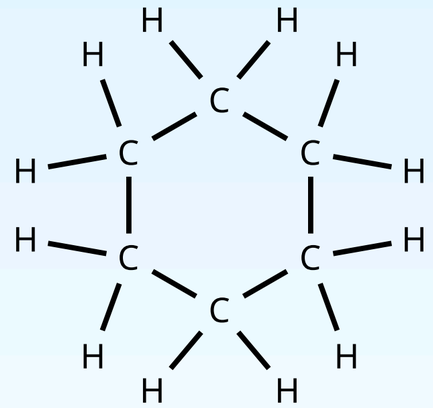
Atom count verified: 4 C, 8 H, 2 O
Valence checked: C=O + ether O, 4 lone pairs total
Same connectivity confirmed: $CH_3C(=O)OCH_2CH_3$

6-CYCLOHEXANE

1.Molecular Formula

C_6H_{12}

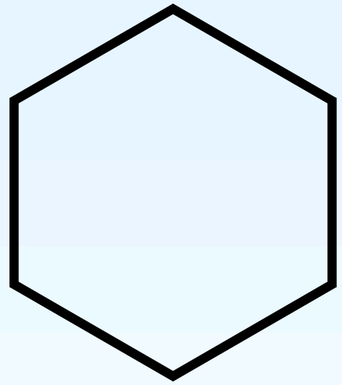
2.Lewis Structure



3.Condensed Structural Formula

$(CH_2)_6$

4.Bond-Line Formula



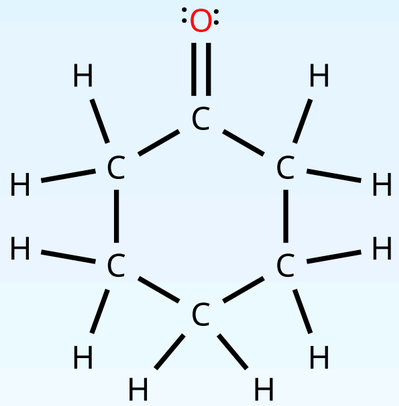
Atom count verified: 6 C, 12 H
Valence checked: each C has 2 ring bonds + 2 H
Same connectivity confirmed: $(CH_2)_6$ ring

7-CYCLOHEXANONE

1.Molecular Formula

$C_6H_{10}O$

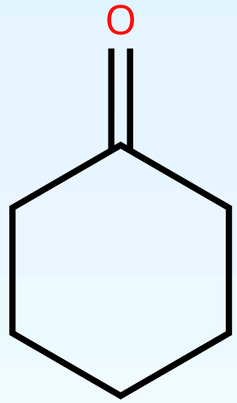
2.Lewis Structure



3.Condensed Structural Formula

$(CH_2)_5C=O$

4.Bond-Line Formula



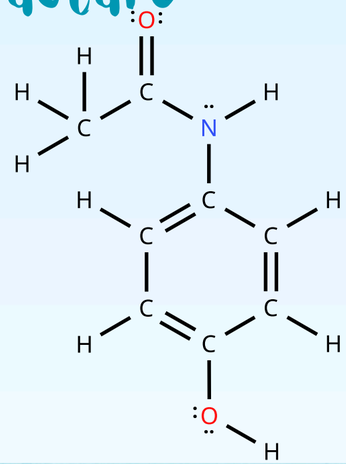
Atom count verified: 6 C, H 10, 1 O
Valence checked: carbonyl C has no H, O has 2 lone pairs
Same connectivity confirmed: $(CH_2)_5C=O$

8-ACETAMINOPHEN (PARACETAMOL)

1.Molecular Formula

$C_8H_9NO_2$

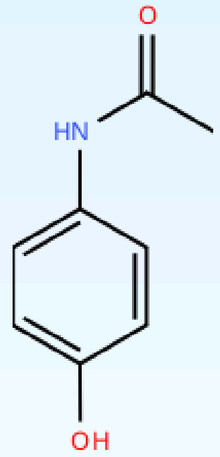
2.Lewis Structure



3.Condensed Structural Formula

$CH_3C(=O)NH\text{-}C_6H_4\text{-}OH$

4.Bond-Line Formula



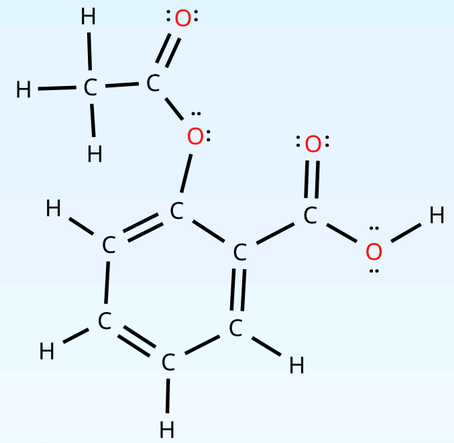
Atom count verified: 8 C, 9 H, 1 N, 2 O
Valence checked: N 3 bonds + 1 lone pair, each O 2 lone pairs
Same connectivity confirmed: NH and OH on opposite ring carbons (para), $CH_3C(=O)NH\text{-}C_6H_4\text{-}OH$

9-ASPIRIN (ACETYLSALICYLIC)

1.Molecular Formula

$C_9H_8O_4$

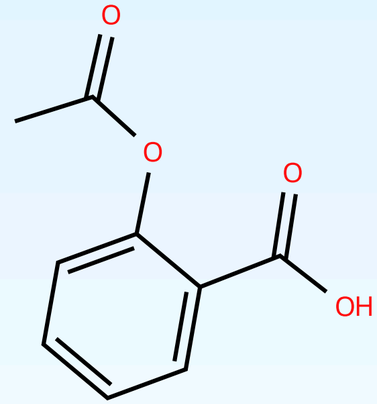
2.Lewis Structure



3.Condensed Structural Formula

$CH_3C(=O)O\text{-}C_6H_4\text{-}C(=O)OH$

4.Bond-Line Formula



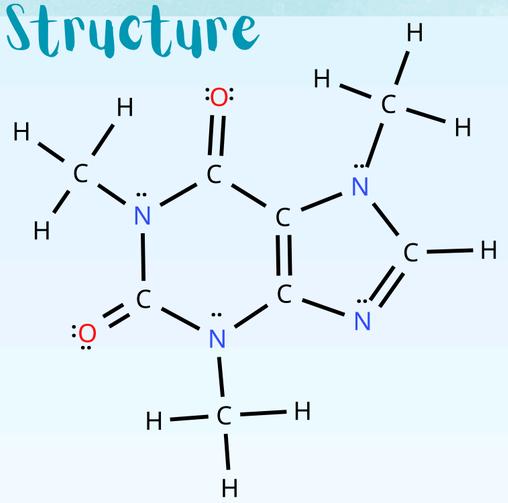
Atom count verified: 9 C, 8 H, 4 O
Valence checked: 2 C=O, each O has 2 lone pairs
Same connectivity confirmed: COOH and $OC(=O)CH_3$ on neighboring ring carbons (ortho), $CH_3C(=O)O\text{-}C_6H_4\text{-}COOH$

10-CAFFEINE

1.Molecular Formula

$C_8H_{10}N_4O_2$

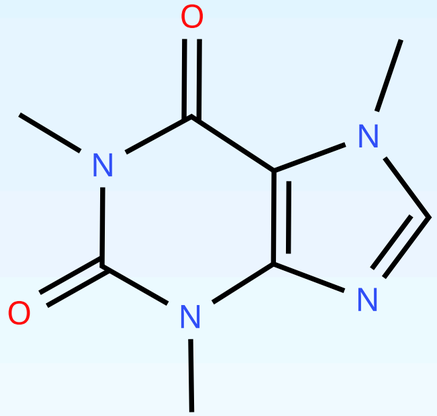
2.Lewis Structure



3.Condensed Structural Formula

$CH_3NC(=O)C_2=C(N=CHN_2CH_3)N(CH_3)C=O$

4.Bond-Line Formula



Atom count verified: 8 C, 10 H, 4 N, 2 O
Valence checked: each N 3 bonds + 1 lone pair, each O 2 lone pairs
Same connectivity confirmed: fudes 6- and 5-ring, 3 N-CH_3, 2 C=O